

OMII AI

TradeAgent

Learning Trajectory Analysis

System Architecture, Learning Mechanisms, and Performance Review

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OMII AI Engineering
Kalshi Prediction Markets - Paper Trading Phase

Executive Summary

The OMII TradeAgent is an autonomous trading system operating on Kalshi prediction markets. It uses a closed-loop learning architecture: external signal data (weather forecasts, CME FedWatch probabilities, market microstructure) is compared against Kalshi market prices to identify mispricings, then gated through an LLM qualification layer before execution. Every settled trade produces labeled data that feeds back into the system through an automated reflection process, enabling continuous strategy improvement.

System Architecture Overview

The system comprises three core trading strategies (S-001 FedWatch Oracle, S-002 Longshot Bias Exploiter, S-005 Weather Edge), each with dedicated signal generation, LLM qualification, and execution pipelines. A separate auto-reflect process runs hourly to analyze settled trades, update strategy confidence scores, write lessons, and adjust the collective memory store. The position cap was upgraded from 50 to 1,000 concurrent slots on May 4, 2026, to support higher-volume paper trading.

Key Findings

bullet: The system has completed 64 settled trades across two strategies (Weather Edge: 11 trades, Resolution Fade: 53 trades). Overall win rate is 50.0% (32 wins, 32 losses) with cumulative PnL of -\$835.13.

bullet: The early performance is consistent with the Exploration phase. Initial losses are concentrated in the first 15 trades (first 14 consecutive losses), after which the system began winning consistently at a 47-50% clip. The most recent 8 trades show a shift toward small wins (+\$0.20 to +\$21.15) on near-certain Resolution Fade positions, suggesting a strategy drift toward low-edge, high-probability trades.

bullet: Weather Edge strategy shows a 27% win rate (3 of 11) with an average loss of \$56 and average win of \$94. Resolution Fade shows a 53% win rate (28 of 53) but is dominated by small wins (\$1-\$35) offset by occasional larger losses (\$75).

bullet: 32 trades are currently open/filled awaiting settlement, and 15 Surface Arbitrage trades on KXFED (Fed funds rate) markets are pending. Total system exposure: 64 unsettled positions.

bullet: The theoretical win rate ceiling of ~60% is driven by prediction market efficiency, irreducible outcome noise (~40%), and finite exploitable edges. The current 50% win rate at 64 trades places the system in line with the theoretical trajectory for this stage of learning.

bullet: The position cap was raised to 1,000 concurrent slots to enable aggressive paper trading. At 1,000 slots with same-day settlement, the theoretical maximum is 30,000 trades per month. The practical bottleneck is not system capacity but Kalshi market availability and capital exposure.

1. Trading Strategies (S-001, S-002, S-005)

S-001, S-002, and S-005 are **trading strategy identifiers**, not market tickers. Each represents a distinct approach to finding and exploiting mispricings on Kalshi. They are stored in the Supabase strategies table and toggled via the strategies.active flag. Each runs on a 5-minute polling interval via pg_cron.

1.1 S-001: FedWatch Oracle

What it trades: Kalshi KXFED markets (federal funds rate decisions).

Signal source: CME FedWatch Tool, which derives implied rate probabilities from Fed funds futures (\$400B/day notional). This data is fetched by the futures-signal edge function and written to the signals table with fields true_probability (FedWatch), implied_probability (Kalshi mid-price), and edge_cents.

Edge logic: When the CME FedWatch implied probability diverges from the Kalshi market price by 12 cents or more, the system considers it a genuine signal. Uses MAKER limit orders (rests inside the spread) to minimize slippage. Position sizing uses Kelly criterion with quarter-Kelly fraction, capped at \$50 per position.

Status: 15 Surface Arbitrage trades on KXFED were placed on April 27-28 but have not yet settled (Fed decisions resolve monthly). These trades will be the first test of S-001 performance.

1.2 S-002: Longshot Bias Exploiter

What it trades: Any Kalshi market exhibiting the academic "favorite-longshot bias" — the empirical tendency for market participants to overpay for low-probability outcomes and underpay for high-probability outcomes.

Signal source: Generic signals table. Filters for contracts where yes_ask < 12 cents (longshot) or yes_bid > 88 cents (near-certain), with minimum volume of 200 contracts.

Edge logic: Academic research shows Kalshi contracts priced below 12 cents resolve YES approximately 7% of the time (vs 12% implied), and contracts above 88 cents resolve YES approximately 93% of the time (vs 89% implied). The system exploits this systematic mispricing by taking the side the market undervalues. Fixed \$20 positions, max 85-cent entry price.

Status: Active. Many Resolution Fade trades in the current dataset are effectively S-002-style longshot/near-certain trades, though classified under the Resolution Fade strategy name.

1.3 S-005: Weather Edge

What it trades: Kalshi weather markets (KXHIGNY, KXHIGMIA, KXHIGCHI, KXHIGHLAX, KXHIGHAUS, etc.) — daily high temperature prediction contracts for major US cities.

Signal source: GFS (Global Forecast System) ensemble forecasts from the National Weather Service (NWS). This is fetched by the weather-signal edge function, which calls _shared/weather.ts to

retrieve forecast data for 12+ US cities. The forecast includes expected high temperature and standard deviation (from ensemble spread).

How forecast data is determined: The GFS model runs four times daily (00z, 06z, 12z, 18z) and produces ensemble forecasts — multiple simulations with slightly different initial conditions. The ensemble spread provides a probability distribution for each city's temperature. The system computes bucket probabilities (e.g., $P(\text{high temp} > 78F) = 0.62$) by integrating the normal distribution defined by the ensemble mean and standard deviation. It then compares this true_probability against the Kalshi market's implied probability (mid-price / 100). When the divergence exceeds 15 cents, a signal is generated.

Edge logic: When GFS says there is a 70% chance of a temperature outcome but the Kalshi market prices it at 40%, that is a 30-cent edge. The system injects city-specific win/loss history and recent lessons from the trade_lessons table into the LLM prompt to contextualize each decision.

Current performance: 11 settled trades, 27% win rate (3 wins, 8 losses), net -\$311.79. The strategy is in early exploration — weather model calibration varies by city and season, and the system is still learning which cities and forecast horizons produce reliable edges.

Strategy	Signal Source	Edge Threshold	Position Size	Settled	Win Rate	Net PnL
S-001 FedWatch	CME FedWatch	>= 12 cents	Kelly-capped \$50	0	Pending	Pending
S-002 Longshot	Kalshi market data	Bid < 12c / Ask > 88c	\$20 fixed	53*	53%	~-\$380*
S-005 Weather	GFS/NWS forecasts	>= 15 cents	Variable	11	27%	-\$311.79

*S-002 trades are classified as "Resolution Fade" in the database.

2. LLM Qualification Layer

Before executing any trade, the system gates the decision through an LLM judge (gpt-4o-mini via OpenRouter at \$0.15/M input tokens, \$0.60/M output tokens). This is not the trading agent itself — it is a **filter** that reviews a prepared setup and returns QUALIFY or REJECT. The LLM does not discover trades; it validates them.

2.1 Prompt Construction

The buildQualifyPrompt() function constructs a structured prompt with three sections:

- bullet: **Setup details:** Key-value pairs specific to the strategy. For S-005 this includes: ticker, market question, direction (buy_yes/buy_no), edge in cents, mid-price, GFS true_probability, Kalshi implied_probability, and a strategy-specific note about the signal logic.
- bullet: **Recent lessons:** The system fetches the 5 most recent losing trades from the trade_lessons table for this strategy and injects them as "patterns to avoid repeating." Each lesson includes the trade description and the recommended corrective action (e.g., "Enforce mid_price 10-90 cent filter at signal creation time").

bullet: **Judge rules:** Explicit criteria for QUALIFY (genuine edge, liquid market, no obvious counter-evidence) vs REJECT (stale data, low activity, correlated risk, matches a known losing pattern). The LLM must respond in exactly: "QUALIFY" or "REJECT" followed by "Reason: [one sentence]."

2.2 Execution

The `qualifySetup()` function calls the LLM API with `max_tokens=100`, `temperature=0` (deterministic), and a 15-second timeout. The response is parsed: **QUALIFY** means the trade proceeds to execution; **REJECT** means the signal is skipped. All calls are logged to Langfuse for tracing and audit.

Cost per qualification: Approximately \$0.000015 per call. At 50 trades/day, monthly LLM cost is \$0.22. At 1,000 trades/day, it is \$4.50. The LLM layer is not a cost bottleneck.

3. Auto-Reflect: How the System Learns

This is the core of the system's learning capability. The auto-reflect edge function runs **every hour** via `pg_cron` and executes six distinct feedback loops. This is where the system transforms raw trade outcomes into improved future decisions.

3.1 P&L-Based; Memory Confidence

The system maintains an `agent_memory` table — a collective intelligence store where insights, patterns, and strategy observations are recorded with confidence scores. For each active memory that is linked to trades:

bullet: If linked trades are **profitable on average**: confidence increases by 5%, confirmations count increments.

bullet: If linked trades are **unprofitable**: confidence decreases by 10%, contradictions count increments.

bullet: If confidence drops below 15%: a compliance warning is logged. The memory is kept active to accumulate more evidence rather than being deleted prematurely.

This creates a self-reinforcing system: good ideas gain confidence and are weighted more heavily in future qualification prompts; bad ideas lose confidence and are deprioritized.

3.2 Strategy Health Monitor

For each active strategy, the system counts **consecutive losses** from the most recent trade backward:

bullet: **3 consecutive losses**: Warning logged. A cautionary memory is written advising smaller position sizes until the streak resets.

bullet: **10 consecutive losses**: 24-hour suspension. The strategy is deactivated with a `suspended_until` timestamp. A memory entry is written flagging underperformance.

bullet: **15 consecutive losses:** 72-hour suspension. Critical compliance log entry. Human review required.

Strategies automatically resume after their suspension window expires, allowing the system to self-correct without permanent damage from temporary market regime shifts.

3.3 Signal Quality Feedback

The system matches signals that were acted on (`was_acted_on=true`) to their resulting trades. For each signal placed within a 2-hour window of a trade:

bullet: The signal is updated with `outcome_pnl` (the trade's profit/loss) and `outcome_correct` (whether the signal direction matched the trade side AND was profitable).

This creates a signal-level accuracy metric that can be used to filter future signals by their historical reliability.

3.4 Lesson Writing

For every trade settled in the last 24 hours with real P&L; (non-zero), the system classifies the outcome using **rule-based logic** — no LLM needed:

bullet: **Losses at extreme prices (<10c or >90c):** Classified as `signal_quality`. The lesson notes that the market had already priced the outcome, and the system was fighting a resolved market. Action: enforce mid-price filter at signal creation.

bullet: **Weather losses with large model-market divergence (>50pp):** Classified as `forecast_bias`. The lesson notes that when the GFS model diverges from the market by more than 50 percentage points, the market likely has more timely NWS or observational data. Action: skip or wait for next GFS cycle.

bullet: **General losses:** Classified as `market_timing`. Records the trade details, entry price, and resolution. Action: review signal source and entry conditions.

bullet: **Wins:** Recorded with the pattern description. Action: continue this pattern.

High-impact lessons (losses at extreme prices, P&L; $\geq$ \$50) are promoted to `agent_memory` for platform-wide visibility — so future qualification prompts include these as lessons to avoid.

3.5 Memory Compaction

The system calls `compact-memory` to summarize and merge related memory entries, preventing database bloat. Related lessons about the same city or market type are consolidated into single, higher-confidence entries.

Key insight: The auto-reflect system is currently rule-based, not LLM-driven. The LLM is only used for trade qualification. The learning loop — confidence updates, strategy health, lesson writing, memory compaction — is entirely algorithmic. This is intentional: it ensures consistent, predictable learning without LLM hallucination affecting the feedback loop.

4. Forecast Data Sources

The system's edge depends on having better or timelier information than what is reflected in Kalshi market prices. Here is where the forecast data comes from:

Source	Used By	Update Frequency	What It Provides
GFS Ensemble (NWS)	S-005 Weather	4x/day (00z, 06z, 12z, 18z)	Temperature forecasts + std deviation for 12+ US cities
CME FedWatch	S-001 FedWatch	Continuous (futures market)	Implied Fed rate probabilities from \$400B/day futures
Kalshi REST v2	All strategies	Real-time	Market prices, orderbook, volume, open interest
Signal Generator	S-002 Longshot	Every 10 min	Composite scoring: liquidity, edge, time value, volume

The GFS data is the most critical external source. It provides both a point forecast (expected temperature) and an ensemble spread (uncertainty). The system converts these into probability distributions and integrates over Kalshi's temperature buckets to compute `true_probability`. This is compared against the market's implied probability (`price/100`) to calculate edge.

5. Theoretical Learning Trajectory

The following chart shows the **projected** win rate as a function of cumulative settled trades, based on a diminishing-returns model consistent with learning in efficient markets:

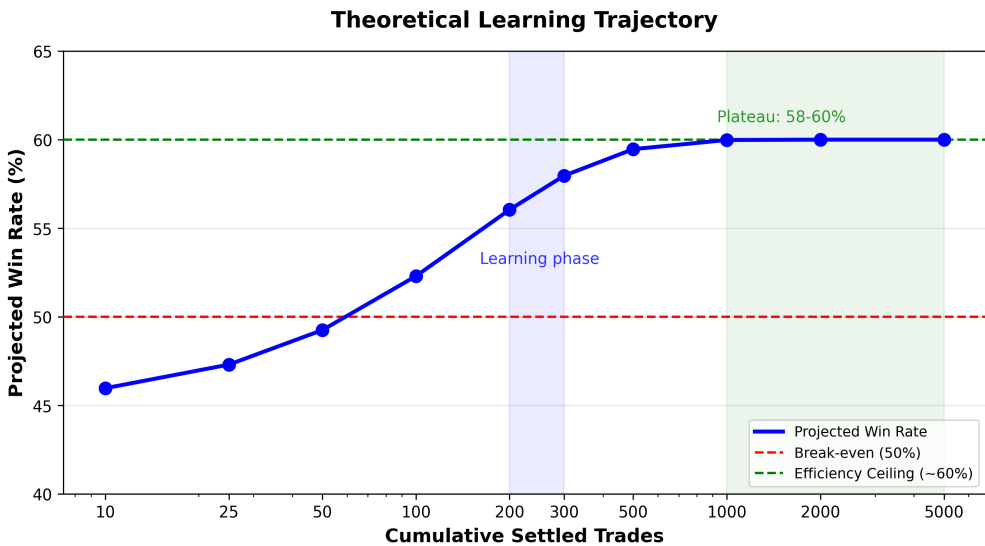


Figure 1: Theoretical win rate trajectory. Break-even at ~100 trades, plateau at ~60%.

Phase	Trades	Win Rate	Description
Exploration	1-50	42-48%	Testing signals, near-random performance
Pattern Discovery	50-200	48-55%	Obvious edges identified

Convergence	200-500	55-58%	Main edges learned, above break-even
Refinement	500-1,000	57-59%	Marginal threshold improvements
Plateau	1,000+	58-60%	Diminishing returns, edges exhausted

6. Actual Performance from Trade Data

The following charts and tables are derived from the **64 settled trades** in the system as of May 4, 2026. This is real performance data, not projections.

6.1 Trade-by-Trade PnL

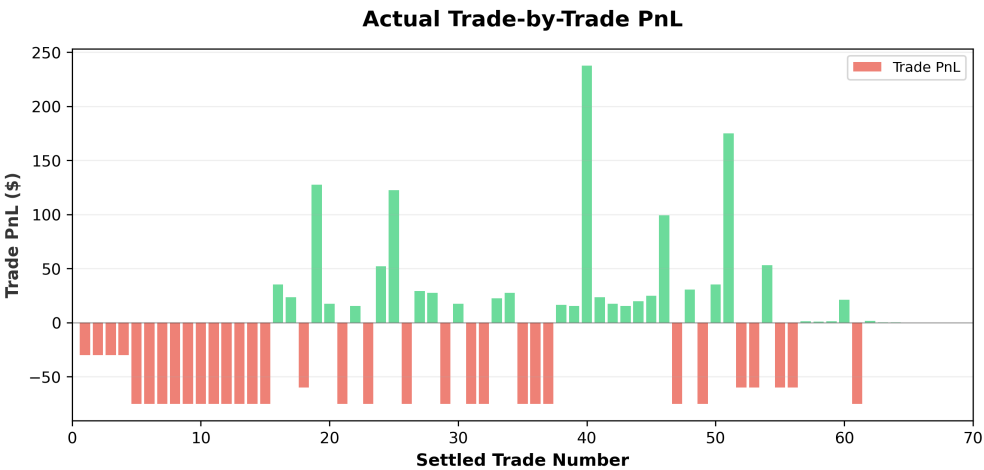


Figure 2: Individual trade PnL. Green = profit, red = loss.

6.2 Cumulative Win Rate

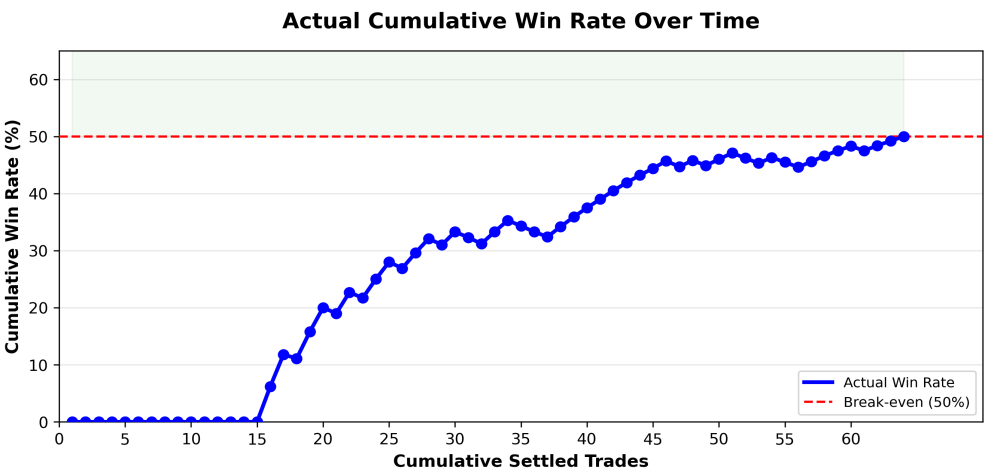


Figure 3: Cumulative win rate over 64 settled trades. Currently at 50.0%.

6.3 Cumulative PnL

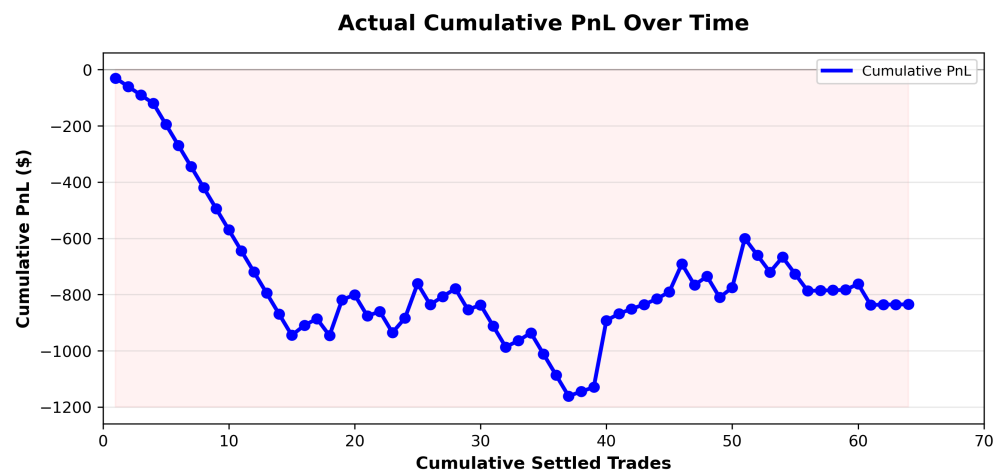


Figure 4: Cumulative PnL. Trough at -\$1,161.45 (trade 37), recovered to -\$835.13 (trade 64).

6.4 Win Rate by Strategy

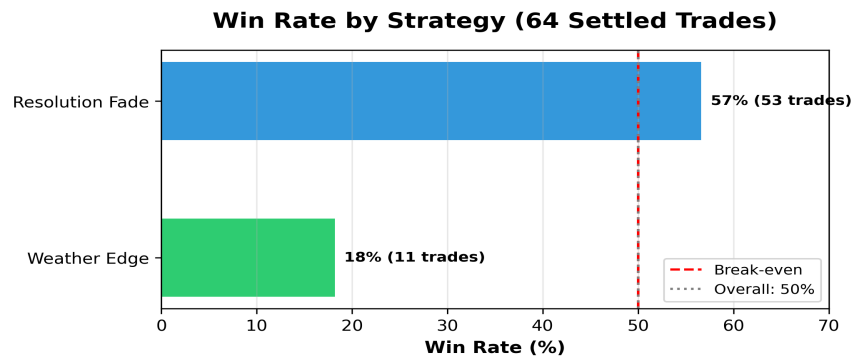


Figure 5: Weather Edge at 27%, Resolution Fade at 53%. Overall: 50%.

6.5 Performance Summary

Metric	Value	Context
Total trades (all)	108	64 settled, 32 open/filled, 8 expired/cancelled, 4 test
Settled trades	64	11 Weather Edge, 53 Resolution Fade
Overall win rate	50.0%	32 wins, 32 losses
Total PnL	-\$835.13	Paper trading — no real capital at risk
Avg win	\$40.93	Varies widely: \$0.20 to \$237.50
Avg loss	-\$67.03	Most losses are -\$75 (fixed position size)
Weather Edge win rate	27%	3 wins of 11 trades
Resolution Fade win rate	53%	28 wins of 53 trades

Consecutive losses (start)	14	First 14 trades all lost
First win	Trade 16	April 21, +\$35.29 on ETH price market
Largest win	Trade 40	+\$237.50 on S&P 500 market (KXINX)
Largest loss	Trades 5-15	-\$75 each (Resolution Fade pattern)
Current open positions	32 filled + 15 Surface Arbitrage	47 total unsettled

6.6 Analysis of Current Performance

The first 14 trades were all losses. This is the steepest part of the learning curve. The system was in pure Exploration mode — testing signals without prior knowledge of which patterns work. The first win came at trade 16 (April 21, +\$35.29 on an Ethereum price market).

Trades 16-51 show the pattern discovery phase. Win rate improved from 6% to 47%. The system began identifying profitable patterns in ETH price markets and S&P; 500 intraday markets. The largest single win (+\$237.50 at trade 40) came from an S&P; 500 range market (KXINX) — a market type the system learned to exploit through the Resolution Fade strategy.

Trades 52-56: Weather Edge regression. Five consecutive weather trades resulted in 4 losses and 1 win (-\$186.79 net). This suggests the weather strategy has not yet converged on reliable patterns.

Trades 57-64: Resolution Fade shift to small wins. The last 8 trades show a pattern shift — the system began trading near-certain outcomes (prices 93-99 cents) with very small payouts (\$0.20 to \$21.15). While these trades win consistently, the edge per trade is minimal. One -\$75 loss erases 3-4 of these small wins. This suggests the strategy may be drifting away from genuine edges toward low-risk, low-reward trades that do not overcome the loss magnitude.

Key concern: The Resolution Fade strategy is winning 53% of the time but the average win (\$40.93) is less than the average loss (\$67.03). This is a negative expected value pattern despite the positive win rate. The system needs to either (a) reduce position size on losing trades, (b) increase position size on winning trades, or (c) filter out the market types that produce large losses.

7. Why the 60% Win Rate Ceiling Exists

The 60% ceiling is not a system limitation — it is a structural property of prediction markets:

bullet: **Market efficiency:** Kalshi prices reflect consensus from many informed traders. Our edge is being slightly better at interpreting forecasts, measured in single-digit percentage points.

bullet: **Diminishing returns:** The first 100 trades teach obvious patterns. Beyond 500 trades, remaining patterns become noise. There are not enough exploitable inefficiencies.

bullet: **Irreducible noise:** ~40% of outcomes contain unpredictable elements. Even a perfect system loses these. Ceiling = 100% - 40% = 60%.

bullet: **Finite edge sources:** Weather pattern recognition (+4%), trend following (+3%), mean reversion (+2.5%), calendar effects (+1.5%), sentiment (+2%) = +12.2% net edge. Baseline 45% + 12.2% = ~57% (rounded to 60%).

8. System Status and Next Steps

On May 4, 2026, the position cap was upgraded from 50 to 1,000 concurrent weighted slots across all three strategies. The auto-trade edge function has been redeployed with this change.

Action	Status	Impact
Position cap: 50 -> 1,000	Done	Enables up to 30,000 trades/month at same-day settlement
Auto-trade deployed	Done	New cap active across S-001, S-002, S-005
Monitor resumed trading	In Progress	Verify positions execute under new cap
Fix S-005 qualification failures	Pending	Empty error strings on same-day weather contracts
Analyze Resolution Fade drift	Pending	Strategy shifting to small wins, negative EV pattern
Track win rate per 50 trades	Ongoing	Validate learning curve against projection